

Homework Assignment #2

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2026-02-13

```
library(readr)
```

Warning: package 'readr' was built under R version 4.4.3

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.1

Warning: package 'lubridate' was built under R version 4.4.1

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v purrr      1.0.2
```

```
v forcats   1.0.0      v stringr    1.5.1
```

```
v ggplot2    4.0.1      v tibble     3.2.1
```

```
v lubridate  1.9.4      v tidyr      1.3.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
```

```
library(clusterProfiler)
```

Warning: package 'clusterProfiler' was built under R version 4.4.2

clusterProfiler v4.14.6 Learn more at <https://yulab-smu.top/contribution-knowledge-mining/>

Please cite:

S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang, W Xie, T Wu, L Xie, G Yu. Using clusterProfiler to characterize multiomics data. Nature Protocols. 2024, 19(11):3292-3320

Attaching package: 'clusterProfiler'

The following object is masked from 'package:purrr':

simplify

The following object is masked from 'package:stats':

filter

```
library(org.Sc.sgd.db)
```

Loading required package: AnnotationDbi

Warning: package 'AnnotationDbi' was built under R version 4.4.1

Loading required package: stats4

Loading required package: BiocGenerics

Warning: package 'BiocGenerics' was built under R version 4.4.1

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:lubridate':

intersect, setdiff, union

The following objects are masked from 'package:dplyr':

combine, intersect, setdiff, union

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
table, tapply, union, unique, unsplit, which.max, which.min

Loading required package: Biobase

Warning: package 'Biobase' was built under R version 4.4.1

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Loading required package: IRanges

Warning: package 'IRanges' was built under R version 4.4.2

Loading required package: S4Vectors

Warning: package 'S4Vectors' was built under R version 4.4.1

Attaching package: 'S4Vectors'

The following object is masked from 'package:clusterProfiler':

rename

The following objects are masked from 'package:lubridate':

```
second, second<-
```

The following objects are masked from 'package:dplyr':

```
first, rename
```

The following object is masked from 'package:tidyr':

```
expand
```

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Attaching package: 'IRanges'

The following object is masked from 'package:clusterProfiler':

```
slice
```

The following object is masked from 'package:lubridate':

```
%within%
```

The following objects are masked from 'package:dplyr':

```
collapse, desc, slice
```

The following object is masked from 'package:purrr':

```
reduce
```

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:clusterProfiler':

```
select
```

The following object is masked from 'package:dplyr':

select

```
library(ggVennDiagram)
```

Warning: package 'ggVennDiagram' was built under R version 4.4.3

Attaching package: 'ggVennDiagram'

The following object is masked from 'package:tidyr':

unite

```
BiocPkgs <- c("fgsea", "clusterProfiler", "AnnotationDbi", "org.Sc.sgd.db")
BiocManager::install(setdiff(BiocPkgs, rownames(installed.packages())))
```

Bioconductor version 3.20 (BiocManager 1.30.27), R 4.4.0 (2024-04-24)

Old packages: 'BART', 'base64enc', 'betareg', 'BH', 'bit', 'blob', 'bmp',
'boot', 'broom', 'bslib', 'car', 'carData', 'class', 'cli', 'clock',
'cluster', 'colorspace', 'cowplot', 'cpp11', 'credentials', 'crosstalk',
'curl', 'data.table', 'dbplyr', 'Deriv', 'digest', 'doBy', 'dplyr', 'dtplyr',
'e1071', 'evaluate', 'fansi', 'fitdistrplus', 'flexdashboard', 'flexmix',
'fontawesome', 'forcats', 'foreign', 'fs', 'future', 'future.apply',
'gargle', 'gbm', 'generics', 'gert', 'GGally', 'ggplot2', 'ggpubr', 'ggsci',
'ggstats', 'gh', 'glmnet', 'globals', 'glue', 'googledrive', 'googlesheets4',
'gplots', 'hardhat', 'haven', 'hms', 'htmltools', 'httr', 'httr2', 'igraph',
'imager', 'irlba', 'isoband', 'jsonlite', 'KernSmooth', 'later', 'lattice',
'lava', 'leidenbase', 'lifecycle', 'listenv', 'lme4', 'lubridate',
'magrittr', 'MASS', 'Matrix', 'MatrixModels', 'mgcv', 'mime', 'modeltools',
'nlme', 'nloptr', 'nnet', 'openssl', 'parallelly', 'patchwork', 'pbkrtest',
'pillar', 'plotly', 'pROC', 'processx', 'prodlim', 'progressr', 'promises',
'proxy', 'ps', 'purrr', 'quantreg', 'R6', 'ragg', 'rappdirs', 'rbibutils',
'Rcpp', 'RcppAnnoy', 'RcppArmadillo', 'Rdpack', 'recipes', 'reformulas',
'reshape2', 'reticulate', 'rlang', 'rmarkdown', 'ROCR', 'RSQLite', 'rstatix',
'rstudioapi', 'rvest', 'sass', 'sctransform', 'selectr', 'Seurat',
'SeuratObject', 'sf', 'shiny', 'sp', 'spam', 'sparsevctrs', 'spatial',
'spatstat.explore', 'spatstat.geom', 'spatstat.random', 'spatstat.univar',
'spatstat.utils', 'stringi', 'stringr', 'survival', 'systemfonts',

```
'textshaping', 'tibble', 'tidyr', 'timechange', 'timeDate', 'tinytex',  
'tree', 'tzdb', 'usethis', 'utf8', 'uuid', 'uwot', 'vctrs', 'viridisLite',  
'vroom', 'wk', 'xgboost', 'xml2', 'yaml', 'yulab.utils', 'zip', 'zoo'
```

```
msn24_EtOH <- readr::read_tsv("https://raw.githubusercontent.com/clstacy/GenomicAnalysis/main/
```

Assignment #1: Due February 17th

Please upload your completed assignments into Blackboard as a .qmd file so that we can reproduce your code when relevant. Save your file as StudentID_Assignment2.

General Rules:

- 1) We strongly encourage people to work in groups and take advantage of any resources at your disposal (including AI—see the syllabus for our complete policy). However, we expect each student to demonstrate their own understanding by generating their own answers and not simply copying (which is plagiarism).
- 2) Science is collaborative, and we encourage students to work through these problems in groups. Please do cite your collaborators if any.
- 3) If you used any outside references, please cite them. Additionally if you used generative AI for the assignment, please provide a link to the prompt and answers. This will help us troubleshoot or provide feedback if the AI makes any mistakes.
- 4) Sonali and I are happy to field questions about the assignments either in class, via email, or by appointment.

Optional DataCamp Assignments:

The class has a free 6-month subscription to DataCamp, which has a whole bunch of interactive courses that span beginner level through advanced. You are free to take any courses of interest for free during the semester, and I would like to point out that there is a sequence of courses for [analyzing genomic data](#) that students may like to use throughout the semester to complement the activities in class.

For this assignment period, I would highly recommend that students complete “[Introduction to Bioconductor in R](#),” Chapters #1 “What is Bioconductor?” and #4 “Introducing Short-Read.”

The Main Assignment:

Using the “02_Gene_Ontology.qmd” exercise as a guide, answer the questions below.

The 4th and 5th code chunks from that qmd file should download a file with all of the gene expression data for the comparison of wild-type yeast (YPS606) and the YPS606 *msn2/4*

mutant, and then change the headers to remove the characters that certain packages don't recognize. The resulting file is named "DE_genes_Msn24_EtOH.txt" and in case you cannot generate that file, it has also been uploaded onto OneDrive under Data Files -> Msn24_EtOH -> Differential_Expression. You can import the file directly into R Studio using the "Import Dataset" in the upper right Environment section. Select "From Text (readr)..." from the dropdown menu, and click "Browse..." to navigate your computer to select the file. Change the delimiter to "Tab," and make sure "First Row as Names" is checked.

If clusterProfiler is working, you'll be able to perform all of the GO enrichment analysis in R. If not, you can still open "DE_genes_Msn24_EtOH.txt" in Excel as a tab delimited text file and then copy and paste gene lists into an online tool. For this exercise, I would recommend the [Princeton GO Term Finder](#).

Notes about the dataset:

There are 5 different strain, environment, or (strain x environment) comparisons in the file:

yps606_wt_et_oh_response: $\log_2(\text{YPS606 ethanol stress} / \text{YPS606 unstressed})$, so positive values denote genes induced by ethanol stress and negative values denote genes repressed by ethanol stress.

yps606_msn2_4dd_et_oh_response: $\log_2(\text{YPS606 } msn2/4 \text{ ethanol stress} / \text{YPS606 } msn2/4 \text{ unstressed})$, and again positive values denote genes induced by ethanol stress and negative values denote genes repressed by ethanol stress.

wt_v_msn2_4dd_et_oh_response: $\log_2((\text{YPS606 } msn2/4 \text{ ethanol stress} / \text{YPS606 } msn2/4 \text{ unstressed}) / (\text{YPS606 ethanol stress} / \text{YPS606 unstressed}))$, so negative values denote genes with lower expression for the *msn2/4* mutant's ethanol response, and positive values denote genes with higher expression for *msn2/4* mutant's ethanol response. E.g., *PGM2* has a logFC during the EtOH response of 7.60 in wild-type YPS606, but only 0.78 for the *msn2/4* mutant's EtOH response. The WT vs. mutant EtOH response comparison yields a logFC of -6.96, consistent with a diminished EtOH response in the mutant.

wt_v_msn2_4dd_unstressed: $\log_2(\text{YPS606 } msn2/4 \text{ unstressed} / \text{YPS606 unstressed})$, so positive values denote genes with higher expression in the *msn2/4* mutant under unstressed conditions, and negative values denote genes with lower expression in the *msn2/4* mutant (note that the genes with the lowest expression in the *msn2/4* mutant are *MSN2* and *MSN4*, which makes sense since those two genes were deleted to create the mutant).

wt_v_msn2_4dd_et_oh_absolute: $\log_2(\text{YPS606 } msn2/4 \text{ ethanol stress} / \text{YPS606 ethanol stress})$, so positive values denote genes with higher expression in the *msn2/4* mutant under ethanol stress, and negative values denote genes with lower expression in the *msn2/4* mutant ethanol stress (note that this is not the same as looking at the EtOH response comparison between strains. While many of the genes with lower expression in the *msn2/4* mutant are induced by EtOH stress in the wild-type strain, this is not universally true (e.g., *MRH1* has a logFC of 0.50 for the wild type ethanol response (not induced), but has -3.09 log₂-fold lower expression in the *msn2/4* mutant for this comparison).

Answer the following questions. Where relevant, either type your answers below each question or include your code. Blank code chunks are included that you can paste from 02_Gene_Ontology.qmd or type commands into:

1. For the YPS606 wild type ethanol response (yps606_wt_et_oh_response), generate two gene lists: one where you select genes with an FDR < 0.01 and that are upregulated with logFC of at least 1, and a second set where you use an FDR < 0.05

```
wtEtOHresponse_up_05 <- msn24_EtOH %>%
  filter(fdr_YPS606_WT_EtOH_response < 0.05,
         logFC_YPS606_WT_etoh_response >= 1)

wtEtOHresponse_up_05_go <- wtEtOHresponse_up_05 %>%
  pull(gene_id) %>%
  enrichGO(
    OrgDb = "org.Sc.sgd.db",
    universe = msn24_EtOH$gene_id,
    keyType = "ORF",
    ont= "BP") %>%
  mutate(richFactor = Count / as.numeric(sub("/\\d+", "", BgRatio))) %>%
  arrange(pvalue)
```

```
wtEtOHresponse_up_01 <- msn24_EtOH %>%
  filter(fdr_YPS606_WT_EtOH_response < 0.01,
         logFC_YPS606_WT_etoh_response >= 1)

wtEtOHresponse_up_01_go <- wtEtOHresponse_up_01 %>%
  pull(gene_id) %>%
  enrichGO(
    OrgDb = "org.Sc.sgd.db",
    universe = msn24_EtOH$gene_id,
    keyType = "ORF",
    ont= "BP") %>%
  mutate(richFactor = Count / as.numeric(sub("/\\d+", "", BgRatio))) %>%
  arrange(pvalue)
```

```
head(wtEtOHresponse_up_01_go)
```

ID	Description	GeneRatio
G0:0005975 G0:0005975	carbohydrate metabolic process	125/1096
G0:0005996 G0:0005996	monosaccharide metabolic process	48/1096
G0:0019362 G0:0019362	pyridine nucleotide metabolic process	44/1096

G0:0046496	G0:0046496	nicotinamide nucleotide metabolic process	43/1096
G0:0019318	G0:0019318	hexose metabolic process	42/1096
G0:0006508	G0:0006508	proteolysis	125/1096

	BgRatio	RichFactor	FoldEnrichment	zScore	pvalue
G0:0005975	247/5544	0.5060729	2.559916	12.448873	9.548454e-29
G0:0005996	86/5544	0.5581395	2.823290	8.458308	9.932879e-14
G0:0019362	79/5544	0.5569620	2.817333	8.075109	1.203594e-12
G0:0046496	78/5544	0.5512821	2.788602	7.896261	3.498011e-12
G0:0019318	76/5544	0.5526316	2.795428	7.822687	5.625077e-12
G0:0006508	366/5544	0.3415301	1.727594	7.148974	1.486900e-11

	p.adjust	qvalue
G0:0005975	1.377842e-25	1.154860e-25
G0:0005996	7.166572e-11	6.006778e-11
G0:0019362	5.789287e-10	4.852385e-10
G0:0046496	1.261907e-09	1.057688e-09
G0:0019318	1.623397e-09	1.360676e-09
G0:0006508	3.575995e-09	2.997278e-09

G0:0005975 YMR105C/YML100W/YKLO35W/YLR258W/YMR196W/YFR053C/YDR001C/YBR120C

G0:0005996

G0:0019362

G0:0046496

G0:0019318

G0:0006508 YOL153C/YLL039C/YBL075C/YLR178C/YNL007C/YBR169C/YNL015W/YBR139W/YBL005W-B/YAL005C

	Count	richFactor
G0:0005975	125	0.5060729
G0:0005996	48	0.5581395
G0:0019362	44	0.5569620
G0:0046496	43	0.5512821
G0:0019318	42	0.5526316
G0:0006508	125	0.3415301

The most significantly enriched GO term at this significance cutoff is “carbohydrate metabolic process”.

2. What percent of the genes would we expect to have that GO term in the DE list under the null hypothesis (and what is the null hypothesis)? What percent of the DE genes actually had that GO term?

For the genes upregulated at $FDR < 0.01$ $\log FC \geq 1$, the background ratio (or the null ratio, what we expect to see if there’s no enrichment of the pathway) of ‘carbohydrate metabolic process’ is 272/5546 or 0.0490444. The gene ratio (the observed ratio) of ‘carbohydrate metabolic

process' is 138/1097 or 0.1257976. This represents an enrichment of ~2.56 (or 256%), indicating that the pathway is highly enriched.

3. For the upregulated comparison in question 1, change the logFC cutoff to at least 0 (so any amount of upregulation, no matter how small). What is the mean and median log fold change for genes with that logFC cutoff and a pval < 0.01 but fdr > 0.01? Are there any GO terms enriched for genes? What was your expectation

```
msn24_EtOH %>%
  filter(fdr_YPS606_WT_EtOH_response > 0.01, # FDR > 0.01
         logFC_YPS606_WT_etoh_response >= 0, # logFC is at least 0
         pvalue_YPS606_WT_etoh_response < 0.01 # pval < 0.01
  ) %>%

  # Get mean and median logFC
  summarise(across(logFC_YPS606_WT_etoh_response,
                    list(mean = mean,
                         median = median),
                    na.rm = TRUE))
```

```
# A tibble: 1 x 2
  logFC_YPS606_WT_etoh_response_mean logFC_YPS606_WT_etoh_response_median
                                <dbl>                                <dbl>
1                                0.583                                0.388
```

4. Create your own novel comparison that was not in the original Gene Ontology exercise. For this novel comparisons, explain biologically why you are interested in this particular comparison and the rationale for the cutoffs chosen for generating a gene list. Paste or enter the code for the novel list in the box below.

```
mutantEtOhresponse <- msn24_EtOH %>%
  filter(fdr_YPS606_msn24dd_EtOH_response < 0.01,
         logFC_YPS606_msn24dd_EtOH_response <= -1
  )

mutantEtOhresponse_go <- mutantEtOhresponse %>%
  pull(gene_id) %>%
  enrichGO(
    OrgDb = "org.Sc.sgd.db",
    universe = msn24_EtOH$gene_id,
    keyType = "ORF",
    ont = "BP") %>%
```

```
mutate(richFactor = Count / as.numeric(sub("/\\d+", "", BgRatio))) %>%
  arrange(pvalue)
```

```
head(mutantEt0hresponse_go)
```

	ID	Description	GeneRatio	BgRatio
G0:0042254	G0:0042254	ribosome biogenesis	299/972	400/5544
G0:0022613	G0:0022613	ribonucleoprotein complex biogenesis	315/972	482/5544
G0:0006364	G0:0006364	rRNA processing	227/972	290/5544
G0:0016072	G0:0016072	rRNA metabolic process	242/972	345/5544
G0:0042274	G0:0042274	ribosomal small subunit biogenesis	118/972	140/5544
G0:0042273	G0:0042273	ribosomal large subunit biogenesis	113/972	131/5544

	RichFactor	FoldEnrichment	zScore	pvalue	p.adjust
G0:0042254	0.7475000	4.263519	31.24051	7.841589e-155	1.169965e-151
G0:0022613	0.6535270	3.727524	28.89239	3.763875e-136	2.807850e-133
G0:0006364	0.7827586	4.464623	27.94232	2.845775e-122	1.415299e-119
G0:0016072	0.7014493	4.000859	26.53675	2.570918e-112	9.589523e-110
G0:0042274	0.8428571	4.807407	21.03721	7.668517e-69	2.288286e-66
G0:0042273	0.8625954	4.919989	20.93408	3.532399e-68	8.783898e-66

	qvalue
G0:0042254	1.088743e-151
G0:0022613	2.612921e-133
G0:0006364	1.317045e-119
G0:0016072	8.923791e-110
G0:0042274	2.129426e-66
G0:0042273	8.174094e-66

G0:0042254
 G0:0022613 YJL050W/YDR101C/YMR049C/YGR158C/YJL069C/YPR010C/YNL132W/YPR144C/YEL055C/YHR197W/Y
 G0:0006364
 G0:0016072
 G0:0042274
 G0:0042273

	Count	richFactor
G0:0042254	299	0.7475000
G0:0022613	315	0.6535270
G0:0006364	227	0.7827586
G0:0016072	242	0.7014493
G0:0042274	118	0.8428571
G0:0042273	113	0.8625954

With this comparison, comparisonEt0hresponse_go, I'm interested in seeing which genes are

significantly up-regulated in the comparison between the WT and the mutant phenotypes.

5. For that novel gene list you chose for yourself, which GO term had the smallest adjusted p-value?

The GO term with the smallest adjusted p-value is ‘proteinogenic amino acid metabolic process’.

6. Generate a list (or lists) and perform GO analyses to answer the following question: of the GO Biological Processes upregulated during the yeast ethanol response, which are likely regulated by the Msn2/4 transcription factors, and which ones are not (list the top 3 or so of each)?

In the `WT.vs.msn24dd_EtOH_response` dataset, negative logFC values indicate that the mutant has a lower response relative to the wildtype, while positive logFC values indicate that the mutant has a greater response relative to the wildtype. So, genes with a positive logFC in this dataset are likely upregulated due to the the Msn2/4 mutant transcription factors.

```
comparisonEtOhresponse <- msn24_EtOH %>%
  filter(fdr_WT.vs.msn24dd_EtOH_response < 0.01,
         logFC_WT.vs.msn24dd_EtOH_response >= 1
  )

comparisonEtOhresponse_go <- comparisonEtOhresponse %>%
  pull(gene_id) %>%
  enrichGO(
    OrgDb = "org.Sc.sgd.db",
    universe = msn24_EtOH$gene_id,
    keyType = "ORF",
    ont= "BP") %>%
  mutate(richFactor = Count / as.numeric(sub("/\\d+", "", BgRatio))) %>%
  arrange(pvalue)

head(comparisonEtOhresponse_go)
```

ID	Description	GeneRatio
G0:0170039 G0:0170039	proteinogenic amino acid metabolic process	16/85
G0:0170033 G0:0170033	L-amino acid metabolic process	16/85
G0:1901605 G0:1901605	alpha-amino acid metabolic process	16/85
G0:0009066 G0:0009066	aspartate family amino acid metabolic process	9/85
G0:0032197 G0:0032197	retrotransposition	7/85
G0:0006520 G0:0006520	amino acid metabolic process	16/85
BgRatio RichFactor FoldEnrichment zScore	pvalue	

G0:0170039	160/5544	0.10000000	6.522353	8.844168	1.539692e-09
G0:0170033	162/5544	0.09876543	6.441830	8.771139	1.851828e-09
G0:1901605	175/5544	0.09142857	5.963294	8.324683	5.773456e-09
G0:0009066	59/5544	0.15254237	9.949352	8.622925	2.049941e-07
G0:0032197	33/5544	0.21212121	13.835294	9.227264	4.801071e-07
G0:0006520	242/5544	0.06611570	4.312299	6.574198	5.605289e-07

	p.adjust	qvalue
G0:0170039	5.527706e-07	4.756273e-07
G0:0170033	5.527706e-07	4.756273e-07
G0:1901605	1.148918e-06	9.885777e-07
G0:0009066	3.059537e-05	2.632556e-05
G0:0032197	3.961001e-05	3.408214e-05
G0:0006520	3.961001e-05	3.408214e-05

G0:0170039	YBL039C/YEL046C/YOL058W/YOR108W/YPR145W/YER091C/YGL009C/YMR189W/YJL200C/YLR180W/Y
G0:0170033	YBL039C/YEL046C/YOL058W/YOR108W/YPR145W/YER091C/YGL009C/YMR189W/YJL200C/YLR180W/Y
G0:1901605	YBL039C/YEL046C/YOL058W/YOR108W/YPR145W/YER091C/YGL009C/YMR189W/YJL200C/YLR180W/Y
G0:0009066	YEL046C/YPR145W/YER091C/Y
G0:0032197	YBL005W-B/YNL284C-B/Y
G0:0006520	YBL039C/YEL046C/YOL058W/YOR108W/YPR145W/YER091C/YGL009C/YMR189W/YJL200C/YLR180W/Y

	Count	richFactor
G0:0170039		